

Practical Session 1

Task 1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that:

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child process using an appropriate `exec()` system call.
4. Allow the parent process to wait for the child using `wait()`.
5. Display the Process ID (PID) of both parent and child processes.

```
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/keyboard_shell.c
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/fork_exec.c
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/fork_exec ls
-bash: ./build/fork_exec: No such file or directory
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/fork_exec.c -o build/fork_exec
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/fork_exec ls
Parent Process Started
Parent PID: 3554

Parent waiting for child...

Child Process Created
Child PID: 3555
Child Parent PID: 3554
Executing command: ls

Makefile  build  include  src

Child process completed.
Parent PID: 3554
Parent process exiting.
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
GNU nano 8.7.1 src/fork_exec.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(int argc, char *argv[]) {
    if (argc < 2) {
        printf("Usage: %s <command>\n", argv[0]);
        return 1;
    }

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("\nChild Process Created\n");
        printf("Child PID: %d\n", getpid());
        printf("Child Parent PID: %d\n", getppid());

        printf("Executing command: %s\n\n", argv[1]);

        execlp(argv[1], argv[1], NULL);

        perror("exec failed");
        exit(1);
    }

    else {
        printf("\nParent waiting for child...\n");

        wait(NULL);

        printf("\nChild process completed.\n");
        printf("Parent PID: %d\n", getpid());
        printf("Parent process exiting.\n");
    }

    return 0;
}
```

Task 2

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
j_rakshitha@RAKSHITHA:~/ossp_project$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             12th Gen Intel(R) Core(TM) i5-12450HX
CPU family:             6
Model:                 151
Thread(s) per core:    2
Core(s) per socket:    6
Socket(s):              1
Stepping:               2
BogoMIPS:               5375.99
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge m
ca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht sysc
all nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xt
opology tsc_reliable nonstop_tsc cpuid tsc_known_freq
pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2
apic movbe popcnt tsc_deadline_timer aes xsave avx f16
c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibr
s ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad
fsgsbase tsc_adjust bmil avx2 smep bmi2 erms invpcid r
dseed adx smap clflushopt clwb sha_ni xsaveopt xsavec
xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vp
clmulqdq rdpid movdiri movdir64b fsrm md_clear seriali
ze ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                    288 KiB (6 instances)
L1i:                    192 KiB (6 instances)
L2:                     7.5 MiB (6 instances)
L3:                     12 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):     0-11
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:             Not affected
Indirect target selection: Not affected
```

```

Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
_rakshitha@RAKSHITHA:~/ossp_project$ free -h
              total        used        free      shared  buff/cache   available
mem:          7.6Gi          576Mi          6.9Gi          3.6Mi          337Mi          7.0Gi
swap:          2.0Gi           0B           2.0Gi
_rakshitha@RAKSHITHA:~/ossp_project$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
_rakshitha@RAKSHITHA:~/ossp_project$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G   0    3.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.9G  4.0K   3.9G   1% /mnt/wsl
drivers         117G 109G   8.4G  93% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.1G  954G   1% /
none            3.9G 132K   3.9G   1% /mnt/wslg
none            3.9G   0    3.9G   0% /usr/lib/wsl/lib
rootfs          3.9G  2.8M   3.8G   1% /init
none            3.9G 524K   3.9G   1% /run
none            3.9G   0    3.9G   0% /run/lock
none            3.9G   0    3.9G   0% /run/shm
none            3.9G  80K   3.9G   1% /mnt/wslg/versions.txt
none            3.9G  80K   3.9G   1% /mnt/wslg/doc
c:\             117G 109G   8.4G  93% /mnt/c
d:\             359G  13G  347G   4% /mnt/d
none            1.0M   0    1.0M   0% /run/credentials/systemd-journald.service

```

```
tmpfs          780M  12K  780M   1% /run/user/1000
j_rakshitha@RAKSHITHA:~/ossp_project$ lsusb
Command 'lsusb' not found, but can be installed with:
sudo apt install usbutils
j_rakshitha@RAKSHITHA:~/ossp_project$ sudo apt install usbutils
[sudo: authenticate] Password:
Installing:
  usbutils

Installing dependencies:
  libusb-1.0-0

Suggested packages:
  usbutils-py

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 40
  Download size: 141 kB
  Space needed: 468 kB / 1024 GB available

Continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libusb-1.0-0 amd64 2:1.0.29-2build1 [56.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 usbutils amd64 1:019-1 [83.9 kB]
Fetched 141 kB in 2s (60.6 kB/s)
Selecting previously unselected package libusb-1.0-0:amd64.
(Reading database ... 43132 files and directories currently installed.)
Preparing to unpack .../libusb-1.0-0_2%3a1.0.29-2build1_amd64.deb ...
Unpacking libusb-1.0-0:amd64 (2:1.0.29-2build1) ...
Selecting previously unselected package usbutils.
Preparing to unpack .../usbutils_1%3a019-1_amd64.deb ...
Unpacking usbutils (1:019-1) ...
Setting up libusb-1.0-0:amd64 (2:1.0.29-2build1) ...
Setting up usbutils (1:019-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
j_rakshitha@RAKSHITHA:~/ossp_project$ lsusb
j_rakshitha@RAKSHITHA:~/ossp_project$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:             12th Gen Intel(R) Core(TM) i5-12450HX
```

```
Model name: 12th Gen Intel(R) Core(TM) i5-12450HX
CPU family: 6
Model: 151
Thread(s) per core: 2
Core(s) per socket: 6
Socket(s): 1
Stepping: 2
BogoMIPS: 5375.99
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge m
ca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht sysc
all nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xt
opology tsc_reliable nonstop_tsc cpuid tsc_known_freq
pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2
apic movbe popcnt tsc_deadline_timer aes xsave avx f16
c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibr
s ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad
fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid r
dseed adx smap clflushopt clwb sha_ni xsaveopt xsavec
xgetbv1 xsaves avx_vnni vnm i umip waitpkg gfni vaes vp
clmulqdq rdpid movdiri movdir64b fsrm md_clear seriali
ze ibt flush_lld arch_capabilities

Virtualization features:
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 288 KiB (6 instances)
L1i: 192 KiB (6 instances)
L2: 7.5 MiB (6 instances)
L3: 12 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-11
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
```

```

Spec rstack overflow:    Not affected
Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prct
                          l
Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointe
                          r sanitization
Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditiona
                          l; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:                  Not affected
Tsa:                    Not affected
Tsx async abort:        Not affected
Vmscape:                Not affected
j_rakshitha@RAKSHITHA:~/ossproject$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.6Gi        584Mi        6.7Gi        3.6Mi        498Mi        7.0Gi
Swap:          2.0Gi          0B          2.0Gi
j_rakshitha@RAKSHITHA:~/ossproject$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk 
sdb   8:16   0 159.4M  1 disk 
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/
j_rakshitha@RAKSHITHA:~/ossproject$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G    0    3.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.9G  4.0K    3.9G   1% /mnt/wsl
drivers         117G 109G    8.4G  93% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.1G  954G   1% /
none            3.9G 132K    3.9G   1% /mnt/wslg
none            3.9G    0    3.9G   0% /usr/lib/wsl/lib
rootfs          3.9G  2.8M    3.8G   1% /init
none            3.9G 532K    3.9G   1% /run
none            3.9G    0    3.9G   0% /run/lock
none            3.9G    0    3.9G   0% /run/shm
none            3.9G   80K    3.9G   1% /mnt/wslg/versions.txt
none            3.9G   80K    3.9G   1% /mnt/wslg/doc
C:\             117G 109G    8.4G  93% /mnt/c
D:\             359G  13G   347G   4% /mnt/d
none            1.0M    0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           3.9G    0    3.9G   0% /tmp
none            1.0M    0    1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M    0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           780M  12K   780M   1% /run/user/1000
j_rakshitha@RAKSHITHA:~/ossproject$ top
top - 05:15:17 up 4:59, 1 user, load average: 0.00, 0.01, 0.00

```

```
j_rakshitha@RAKSHITHA:~/ossp_project$ top
```

```
top - 05:15:29 up 4:59, 1 user, load average: 0.00, 0.01, 0.00
```

```
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
```

```
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

```
MiB Mem : 7794.6 total, 7188.5 free, 591.8 used, 171.2 buff/cache
```

```
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7202.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24252	15412	11556	S	0.0	0.2	0:01.88	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.14	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50404	17112	15800	S	0.0	0.2	0:00.78	systemd-journal
82	systemd+	20	0	22416	14484	11988	S	0.0	0.2	0:00.15	systemd-resolve
92	root	20	0	35380	12332	9228	S	0.0	0.2	0:06.36	systemd-udev
108	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.06	chronyd-starter
109	root	20	0	4472	3120	2872	S	0.0	0.0	0:00.05	cron
110	message+	20	0	8816	5424	4712	S	0.0	0.1	0:00.22	dbus-daemon
157	root	20	0	44096	29816	14676	S	0.0	0.4	0:00.29	networkd-dispat
168	root	20	0	18436	9416	8168	S	0.0	0.1	0:00.20	systemd-logind
201	syslog	20	0	220548	5776	4464	S	0.0	0.1	0:00.18	rsyslogd
238	root	20	0	5492	2696	2508	S	0.0	0.0	0:00.00	agetty
242	root	20	0	123460	32700	18092	S	0.0	0.4	0:00.21	unattended-upgr
272	_chrony	20	0	23820	11388	7212	S	0.0	0.1	0:00.67	chronyd
282	_chrony	20	0	12096	2468	1820	S	0.0	0.0	0:00.04	chronyd
329	root	20	0	8652	5548	4704	S	0.0	0.1	0:00.01	login
383	j_raksh+	20	0	22340	13584	10968	S	0.0	0.2	0:00.18	systemd
385	j_raksh+	20	0	22912	4108	2108	S	0.0	0.1	0:00.00	(sd-pam)
413	j_raksh+	20	0	6136	5484	3928	S	0.0	0.1	0:00.04	bash
725	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader
726	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.01	Relay(733)
733	j_raksh+	20	0	6268	5548	3872	S	0.0	0.1	0:00.05	bash
939	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader
940	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.69	Relay(941)
941	j_raksh+	20	0	6268	5544	3872	S	0.0	0.1	0:00.30	bash
3755	root	20	0	340148	22240	19180	S	0.0	0.3	0:00.07	packagekitd
3760	polkitd	20	0	306444	8472	7660	S	0.0	0.1	0:00.10	polkitd
3852	j_raksh+	20	0	8180	6020	3900	R	0.0	0.1	0:00.09	top